

Design and Modeling of Differential Capacitive Hexagonal Beam based MEMS Accelerometer

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Abstract—In this study, we designed a differential capacitive-based MicroElectro-Mechanical System (MEMS) accelerometer with hexagonal suspension springs. We utilized standard silicon-on-insulator (SOI) technology with a 30 μm structural layer thickness. The proposed device was modeled using the MEMS+ numerical tool to characterize various performance characteristics. The MEMS accelerometer exhibits an in-plane mode natural frequency of 874.96 Hz, with mechanical and capacitive sensitivities of 0.324 $\mu\text{m}/\text{g}$ and 0.321 pF/g for an input acceleration of $\pm 5\text{g}$, respectively. Furthermore, we investigated the influence of hexagonal beam width on the natural frequency, quality factor, mechanical sensitivity, capacitive sensitivity, bandwidth, and pull-in voltage. It was observed that the choice of beam width significantly influences the performance characteristics of the proposed device. Further, a comparison was made with the accelerometer utilizing an equivalent uniform rectangular suspension spring. From the obtained results, it has been observed that the accelerometer with hexagonal suspension springs provides greater flexibility in measuring low-g applications. Hence, this work provides valuable insights for optimizing future MEMS accelerometer designs to meet the diverse application requirements.

Index Terms—MEMS accelerometer, Differential Capacitance, Hexagonal Microbeam, Natural Frequency, Displacement sensitivity, Capacitive sensitivity, Bandwidth, Pull-in voltage.

I. INTRODUCTION

THE Micro-Electro-Mechanical Systems (MEMS) accelerometer is a type of sensor that is used to measure acceleration. It is a miniature device that contains tiny mechanical structures and electronic components integrated onto a silicon chip [1]. MEMS accelerometers are widely used in various applications, including smartphones, automotive systems, robotics, aerospace, and more, for measuring acceleration and detecting changes in motion during landing [2]. MEMS accelerometers operate on the principle of capacitance change caused by the movement of a proof mass in response to external acceleration forces. By measuring the change in capacitance, these sensors can accurately detect and quantify acceleration along one or more axes, making them essential components in a wide range of applications. MEMS accelerometers have gained widespread adoption across different industries owing to their compact size, enhanced performance characteristics, and seamless integration capabilities with readout circuits for sensing purposes [3]. Designing

MEMS accelerometers involves careful consideration of crucial parameters such as natural frequency, quality factor, pull-in voltage, bandwidth, linearity, bias stability, and resolution. Achieving superior performance necessitates optimizing the device for higher resolution and stability.

In recent years, significant research has been devoted to enhancing the design and optimization of MEMS accelerometers to achieve superior performance and accuracy. Veena et al., [4] employed two symmetric proof masses supported by a single folded beam at the centre and determined the displacement sensitivity to be 0.035362 $\mu\text{m}/\text{g}$. Zhang et al., [5] presented the analytical modeling, simulation, and experimental validation of sensitivity, temperature variation, and active controllability of MEMS geometric anti-spring (GAS) MEMS accelerometer. Sanyal et al., [6] designed a MEMS Capacitive Accelerometer with a symmetrical beam-mass structure and different interconnections, detailing the design of an H-shaped beam capacitive accelerometer using finite element analysis software, analyzing output voltage relative to given accelerations, and proposed two new connector structures to enhance sensitivity, resulting in a significant increase in displacement of the proof mass and sensitivity of the device. Li et al., [7] detailed the design optimization, numerical modeling, fabrication, and characterization of a compact single-axis comb-drive MEMS gap-changeable differential capacitive accelerometer using serpentine beams, achieving high sensitivity of 80 fF/g through optimized gap ratios. Mukiya et al. developed a highly sensitive single-axis in-plane MEMS differential capacitive accelerometer with folded suspension springs using DRIE-based SOI wafer technology. It features analytical modeling, system-level simulations, and integration with readout electronics, achieving a displacement sensitivity of 0.121 $\mu\text{m}/\text{g}$, capacitive sensitivity of 225 fF/g within an operating range of $\pm 5\text{g}$, and a bandwidth of 100 Hz. Recently, Jin et al., [9] introduced a high-performance MEMS accelerometer incorporating DC/AC electrostatic stiffness tuning, featuring a dynamical model enabling self-centring closed-loop operation to nullify temperature drift, stiffness closed-loop to prevent pull-in instability, and real-time adjustments for temperature compensation, achieving a low-temperature drift coefficient (TDC) of approximately 7 $\mu\text{g}/^\circ\text{C}$ and an Allan bias instability of less than 1 μg . Sthuthi et al., [10] have

developed a design, modeling, and simulation of a dual-axis crab-legged suspension system for a capacitive-based MEMS accelerometer, utilizing the SOI MUMPs fabrication process, aiming to achieve specific design characteristics such as natural frequency, cross-axis sensitivity, frequency response, and static capacitance analysis, with simulation conducted in COMSOL Multiphysics. Recently, Wang et al., [11] performed the theoretical and experimental development of a customizable microstructured hexagonal cantilever probe on an optical fiber tip for microforce detection, utilizing femtosecond laser two-photon polymerization (TPP) technology, with the ability to actively tune the elastic constant and mechanical properties, exhibiting promising force sensitivity, Q-factor, and detection limit. Additionally, in a previous study [12] we conducted both numerical simulations and experimental investigations to comprehensively analyze the frequency tuning and damping characteristics of hexagonally shaped microcantilever beams.

Hence, based on the comprehensive review of existing literature, it is apparent that much of the research within the field of MEMS accelerometer optimization has focussed on configurations employing conventional beams such as uniform, folded, and geometric anti-spring-based curved beams. However, there remains a notable gap in research concerning beams with perforations. Therefore, this study introduces a novel design approach for MEMS accelerometers utilizing hexagonal microbeams as suspension springs. The investigation involves conducting a series of numerical simulations using the MEMS+ numerical tool to comprehensively evaluate the proposed design's performance characteristics. Through this work, we aim to contribute to the advancement of MEMS accelerometer technology by exploring innovative design configurations and assessing their potential applications in various fields.

The paper is structured as follows: Section II presents the fundamental mathematical modeling of the MEMS accelerometer employing a spring-mass-damper system. In Section III, we provide a comprehensive overview of the design of the silicon-on-insulator (SOI) based MEMS accelerometer incorporating hexagonal suspension springs. Following this, Section IV presents the findings and discussions corresponding to the different performance characteristics, while Section V concludes the paper.

II. MATHEMATICAL MODELING

In this section, we discuss the basic mathematical modeling of MEMS accelerometer using a spring-mass-damper system.

A. Frequency Equation

Let us consider the MEMS accelerometer shown in figure.1 as a spring-mass-damper system. From the force equilibrium, the governing equation of motion can be found as,

$$M\ddot{y} + C\dot{y} + Ky = Ma_e \quad (1)$$

where, M is the mass of the proof mass, C is the damping coefficient, K is the effective spring constant or stiffness,

a_e external applied acceleration. Here, $\ddot{y}$, represents the time derivative of the displacement of the proof mass.

By taking Laplace transform on both sides of the equation (1), the transfer function of the spring-mass-damper system can be obtained as,

$$TF = \frac{y(s)}{a_e(s)} = \frac{1}{Ms^2 + Cs + K} = \frac{1}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (2)$$

where, ζ is called the damping ratio and ω_n angular natural frequency. By comparing the equation (1) and (2), we can write natural frequency f_n as ,

$$\omega_n = \sqrt{\frac{K}{M}} \rightarrow f_n = \frac{1}{2\pi} \sqrt{\frac{K}{M}} \quad (3)$$

Similarly,

$$2\zeta\omega_n = \frac{C}{M} \rightarrow \zeta = \frac{C}{2M\omega_n}. \quad (4)$$

Under the small damping assumption, the quality factor (Q) is defined as,

$$Q = \frac{1}{2\zeta} = \frac{M\omega_n}{C} \quad (5)$$

Further, under the static condition (all the time derivative terms go zero in equation 6) of the MEMS accelerometer,

$$\frac{y}{a_e} = \frac{M}{K} = \frac{1}{\omega_n^2} \quad (6)$$

From the above equation, it is evident that the sensitivity (output/input) is inversely proportional to the square of the natural frequency of the system.

III. DESIGN OF ACCELEROMETER

In this section, we discuss the design and modeling methodologies of the proposed MEMS accelerometer. We utilized the MEMS+ software to simulate the fabrication process of silicon-on-insulator (SOI) technology for designing the accelerometer. The chosen material has Young's modulus (E), Poisson's ratio (ν), and density (ρ) as 167Gpa, 0.278, and 2329kg/m³, respectively. Following this, the fabrication process of the MEMS accelerometer, initiated from the SOI wafer, progresses to the release of the free-standing structure, which is designed using MEMS+ software. Figure.1 shows the schematic of the designed MEMS accelerometer with various components, and the corresponding parameters used are shown in Table.I.

IV. RESULTS AND DISCUSSION

In this section, we have a comprehensive discussion of the different analyses conducted on the designed MEMS accelerometer employing MEMS+ software. Our study includes a thorough examination of the various aspects and performance characteristics of the accelerometer through the application of different solvers and simulations within the MEMS+ software platform.

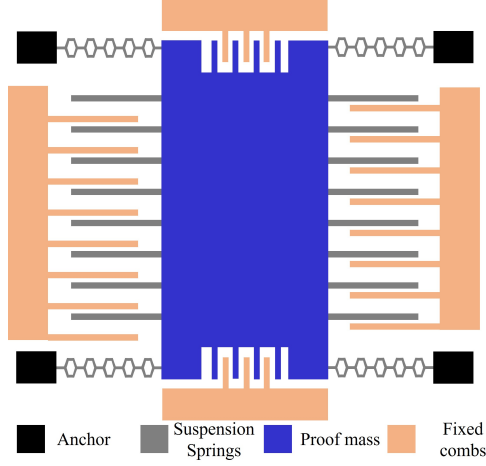


Fig. 1: Schematic of hexagonal beam MEMS accelerometer.

TABLE I: Parameters of MEMS accelerometer

Parameter	Value
Proof mass dimensions ($L_{PM} \times W_{PM}$)	$5778 \mu m \times 2311 \mu m$
Thickness (t_s)	$30 \mu m$
Width of the connecting beam (b)	$5 \mu m$
Length of the comb (l_c)	$300 \mu m$
Width of the comb (w_c)	$5 \mu m$
Overlap length (l_o)	$200 \mu m$
Finger spacing (d_1)	$5 \mu m$
Anti finger spacing (d_2)	$10 \mu m$
Number of comb pairs (N_c)	231
Length of the suspension beam (L_b)	$557 \mu m$
width of beam (w_b)	$35 \mu m$
Mass of the proof mass (M)	0.074665 mg
Effective Stiffness (K)	23.6768 N/m
Relative permittivity of air (ϵ_r)	1
Permittivity of free space (ϵ_0)	$8.85418782 \times 10^{-12} \text{ F/m}$
Acceleration due to gravity (g)	9.81 m/sec^2

A. Modal and Pull-in analysis

In the initial stages of our investigation into MEMS accelerometers, we performed modal analysis using the Sparse Modal Solver within the MEMS+ numerical tool. This numerical process comprised a maximum of 300 iterations, ensuring robust computational convergence with a tolerance set at $1e-15$. The resulting mode shape and the associated in-plane mode's natural frequency for the accelerometer device are depicted in figure 2. Our modal analysis revealed different modes of oscillations along the in-plane (Y-axis), out-of-plane (Z-axis), and torsional (across X-axis) modes, characterized by natural frequencies of 874.96 Hz, 1479.05 Hz, and 2552.75 Hz, respectively. The comprehensive investigation led us to identify the in-plane mode (Y-axis) as the predominant mode of oscillation for the MEMS accelerometer. This highlights the importance of the in-plane mode in influencing the fundamental operational characteristics of the accelerometer device.

Further, we examine the parametric variation of the beam width (w_b) to assess its impact on key performance characteristics. Specifically, we examine the effects on natural frequency, quality factor, displacement sensitivity, capacitance sensitivity,

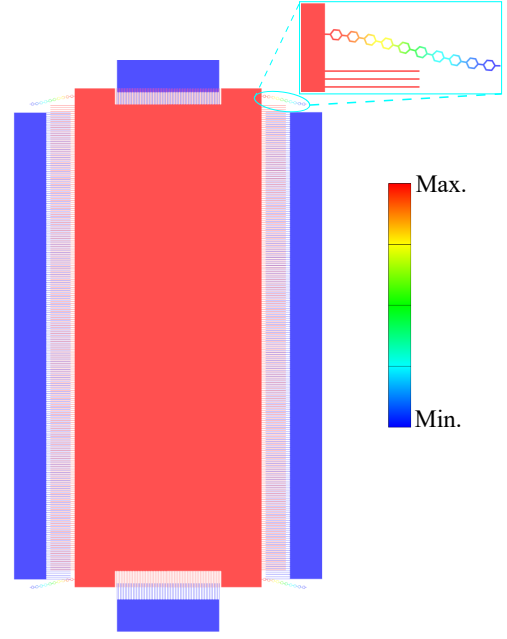


Fig. 2: The mode shape associated with the in-plane mode of a MEMS accelerometer, featuring a hexagonal beam as its suspension spring, exhibits a frequency of 874.263 Hz.

bandwidth, and pull-in voltage. Additionally, we performed the characterization of the quality factor (Q) and stiffness (K) corresponding to the in-plane mode of the MEMS device, as illustrated in Figure 3(a). The outcomes reveal a clear trend wherein the quality factor and stiffness increase with the increase in beam width. Notably, there is a substantial 124.4% increase in the quality factor when the beam width changes from $35 \mu m$ to $155 \mu m$. This considerable enhancement in the quality factor is attributed to a corresponding increase in the stiffness of the MEMS device. Subsequently, we also conducted the pull-in analysis of the designed device in the MEMS+ numerical tool to calculate pull-in and release voltages and understand the behaviour of the device both within and beyond these voltage thresholds. In the design process of MEMS accelerometers, the pull-in voltage is a critical parameter, defined as the voltage at which an imbalance occurs between the electrostatic and mechanical restoring forces. This imbalance leads to the rotor combs snapping through the fixed combs of the MEMS accelerometer. To determine these pull-in voltages (V_{pi}) for varying beam widths, we employed the Pull-in solver, utilizing the arc length continuation algorithm within the MEMS+ numerical tool, as depicted in figure 3(b). The numerical results from the analysis show that the pull-in voltage (V_{pi}) increases with the widening of the beams. This behaviour can be attributed to the increased lateral stiffness of the suspension springs at higher widths. Hence, the observed correlation between beam width, stiffness, quality factor and pull-in voltage holds implications for optimizing the performance of the MEMS accelerometer, indicating that tuning the beam width can be a strategic parameter for improving the

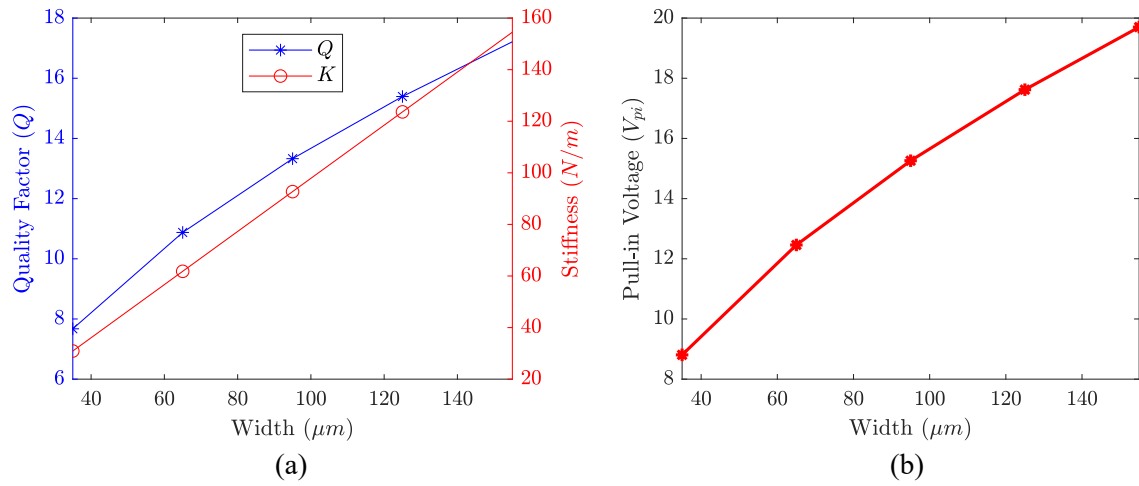


Fig. 3: Influence of beam width variation over (a) Quality factor, Stiffness associated with the in-plane mode of vibration, and (b) Pull-in voltage (V_{Pi}) of MEMS accelerometer, respectively.

dynamic characteristics of the device.

B. Frequency tuning and Electromechanical modeling

1) *Frequency tuning*: Figure 4(a) shows a MEMS accelerometer featuring a hexagonal suspension spring with variable beam width, denoted as w_b . Each individual hexagonal suspension spring has a set width of $35 \mu m$, and these hexagon springs are linked at their contact points to form hexagonal mat configurations, as depicted in Figure 4(a). The widths of these configurations range from $35 \mu m$ to $155 \mu m$. In figure 4(b), the influence of hexagon spring width (w_b) on the in-plane (f_1), out-of-plane (f_2), and torsional mode along X-axis (f_3) of the MEMS accelerometer is illustrated. The results demonstrate a direct correlation between the natural frequencies and the beam width, ranging from 35 to $155 \mu m$. As depicted in figure 4(b), an increase in beam width leads to higher natural frequencies, and the modes become more distinct with the widening of the beam. This observation suggests an enhanced device performance, as it ensures the separation of spurious modes from the desired in-plane operational mode of the MEMS accelerometer.

2) *static and electromechanical analysis*: Further, we explored the static and electromechanical analysis of the MEMS accelerometer to find the displacement and capacitance sensitivity for varying beam widths. Figure 4(c) shows the maximum displacement experienced by the proof mass under the influence of input acceleration of $\pm 5g$ with varying beam width. The results have shown that the maximum displacement of proof mass decreases as the beam width changes from 35 to $155 \mu m$; this is due to an increase in the stiffness of the beams, which resists the motion of the proof mass to the external input acceleration. The mechanical or displacement sensitivity characteristics of the MEMS accelerometer are presented in Figure 5(a), considering a beam width range from 35 to $155 \mu m$. Static analysis reveals a notable trend wherein an increase

in beam width corresponds to a decrease in displacement sensitivity. This can be attributed to the increased stiffness of the suspension springs as the beam width increases. Additionally, our study identified the maximum displacement sensitivity at $0.323 \mu m/g$ for a beam width of $35 \mu m$. A substantial 79.88% reduction in displacement sensitivity was observed when transitioning from a single hexagonal suspension spring to a hexagonal mat of $155 \mu m$ beam width. This finding shows the impact of beam width on the accelerometer displacement characteristics and offers valuable insights for optimizing its mechanical sensitivity for better device performance.

Furthermore, our study extended to finding the differential change in capacitance (ΔC) in response to an input acceleration range of $\pm 5g$ for various beam widths, as illustrated in figure 4(d). Similar to the displacement characteristics observed earlier, the magnitude of ΔC reduces with the beam width varying from 35 to $155 \mu m$. The maximum ΔC value is 1.814 pF under a $5g$ input acceleration is found for a single hexagonal suspension spring of $35 \mu m$. A substantial 463.35% increase in ΔC is noted when comparing a device with a hexagonal mat of $155 \mu m$ width to its counterpart. Figure 5(b) illustrates the capacitance sensitivity (S_c in pF/g) of the designed device, providing valuable insights into its electromechanical behaviour. Analysis of the results reveals a consistent trend where the capacitance sensitivity decreases with the widening of the beam. This behaviour is attributed to the increased stiffness of the beams at larger widths. Notably, the maximum value of S_c is found at 0.321 pF/g for a single hexagonal suspension spring with a width of $35 \mu m$. This value also represents a substantial 400.63% increase in S_c compared to the accelerometer with a hexagonal mat of $155 \mu m$ beam width. Hence, This finding provides a basic insight into the significant influence of beam width on the capacitance response of the device, providing valuable understanding for optimizing its electromechanical performance under the influ-

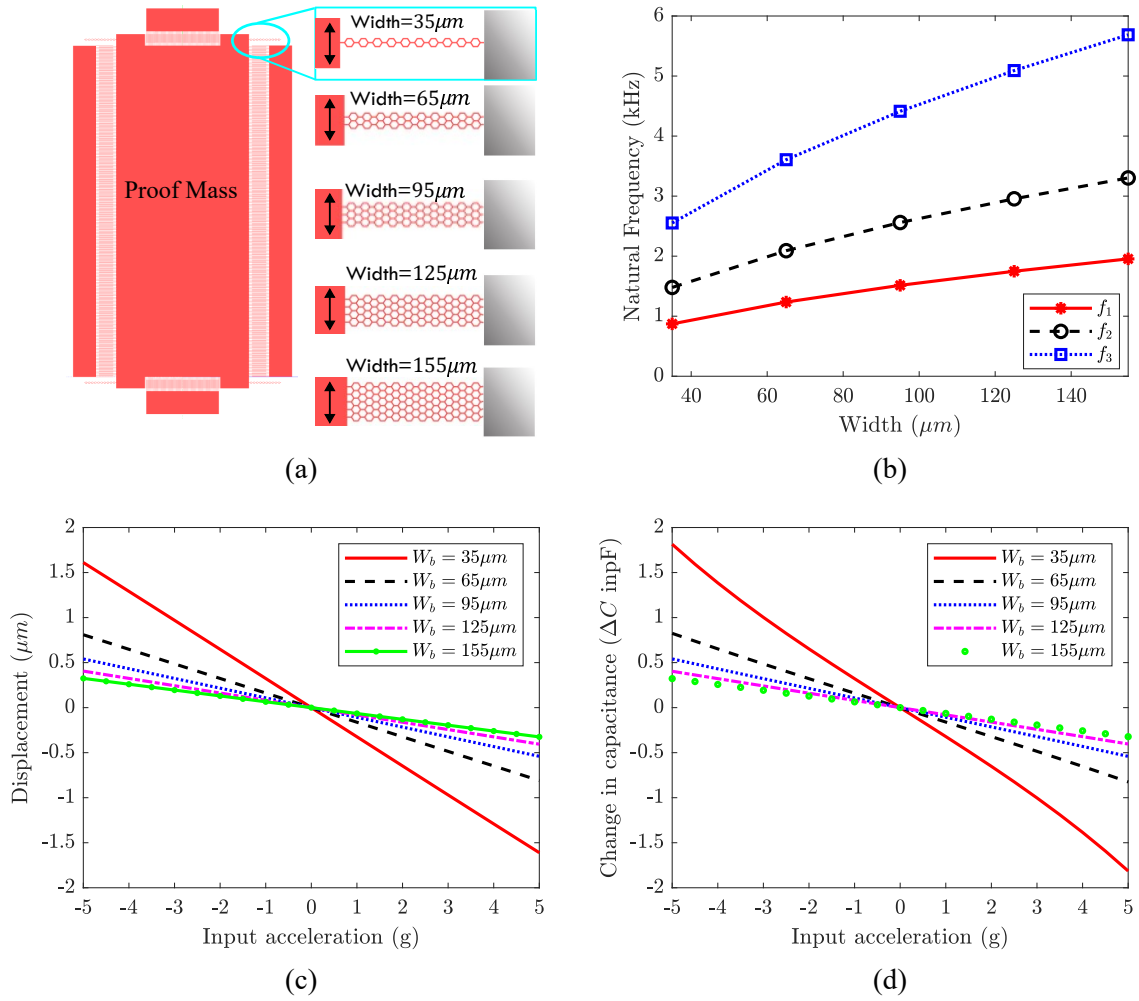


Fig. 4: (a) MEMS accelerometer with different beam configurations with varying width of the beam, comparison of (b) In-plane, out-of-plane and torsional frequency of hexagonal beam (HB) for the length ($L_b = 557 \mu m$) and width ($W_b = 35 \mu m$), and (c) Maximum displacement experienced by proof mass, and (d) change in capacitance (ΔC) of the MEMS accelerometer under the influence of input acceleration of $\pm 5g$ along Y-direction, respectively.

ence of external input accelerations.

3) *Bandwidth analysis:* The bandwidth analysis of the designed device was conducted in the MEMS+ software interface with MATLAB scripting. The bandwidth of an accelerometer is defined as the frequency range within which the magnitude response remains within a specified maximum allowable deviation ($\pm 3dB$) relative to the DC (or low-frequency) value of the magnitude response. Figure.5 (c) shows the influence of beam width variation over the bandwidth measurement of the MEMS accelerometer. As the width of the beam increases, the corresponding bandwidth of the devices increases due to an increase in the stiffness of beams at higher widths. However, it is crucial to note that despite this increased bandwidth at higher beam widths, the magnitude of the frequency response of the device is not as significant. Furthermore, there is a trade-off, as the sensitivity of the device decreases at higher beam widths. This implies that while widening the beam

enhances the device's bandwidth, it concurrently compromises its sensitivity, indicating the necessary balance needed for optimal performance in sensing input accelerations.

C. Comparison with MEMS accelerometer with uniform rectangular suspension spring

A comparative analysis was conducted to evaluate the performance characteristics of MEMS accelerometers employing hexagonal and equivalent uniform rectangular suspension springs. Parameters including natural frequency, quality factor, displacement sensitivity, capacitance sensitivity, bandwidth, and pull-in voltage were analysed. The results revealed distinctive trends between the two suspension designs. Hexagonal suspension springs demonstrated enhanced performance in certain aspects, such as stiffness, natural frequency and displacement sensitivity, and capacitance sensitivity, attributed to their geometric configuration optimizing stiffness. Conversely,

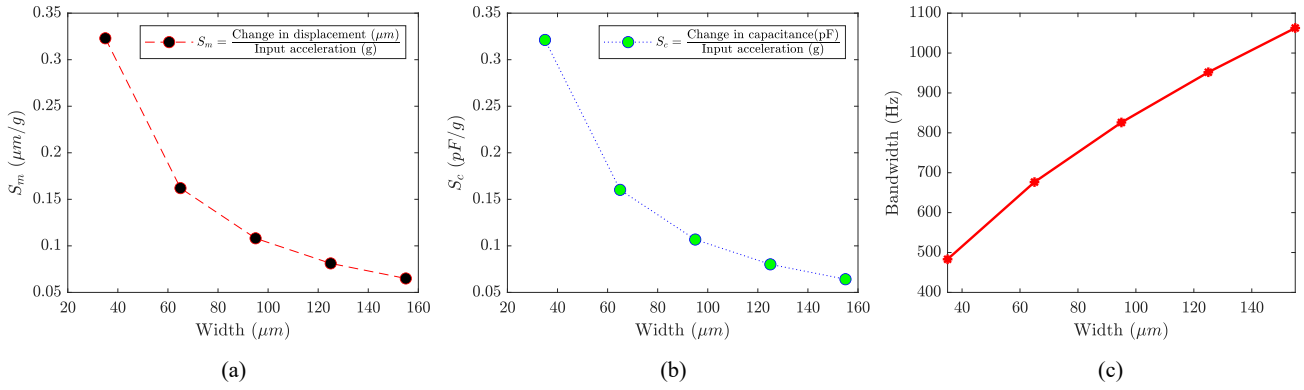


Fig. 5: (a) Displacement sensitivity (S_m) vs beam width, (b) Capacitance sensitivity (S_c) vs beam width. and (c) Bandwidth (Hz) vs beam width.

Suspension spring	Dimensions (μm)	K (N/m)	Q	V_{pi} (V)	f_1 (kHz)	S_m ($\mu m/g$)	S_c (pF/g)	BW (Hz)
Uniform beam	$w_b = 35$; $L_b = 557$	5031.61	92.09	112.18	11.171	0.00199	0.00196	6042.43
Hexagonal beam	$w_b = 35$; $L_b = 557$	30.91	7.69	8.81	0.874	0.32374	0.32122	482.99

TABLE II: Comparison of the performance characteristics between a MEMS accelerometer supported with single hexagonal suspension springs and one with equivalent uniform rectangular suspension springs using MEMS+ numerical tool.

uniform rectangular suspension springs exhibited advantages in other areas, such as quality factor, pull-in voltage and bandwidth. These findings provide the importance of considering suspension spring geometry in MEMS accelerometer design to achieve desired performance characteristics, with potential trade-offs between different parameters depending on the specific application requirements.

V. CONCLUSIONS

In this study, we have designed a differential capacitive-based MEMS accelerometer incorporating hexagonal suspension springs using standard silicon-on-insulator (SOI) technology with a structural layer thickness of 30 μm . Through comprehensive modeling using the MEMS+ numerical tool, we have characterized various performance characteristics of the proposed device. Our results demonstrate an in-plane mode natural frequency of 874.96 Hz, accompanied by mechanical and capacitive sensitivities of 0.324 $\mu m/g$ and 0.321 pF/g, respectively, for an input acceleration of $\pm 5g$. Additionally, we have investigated the impact of hexagonal beam width on key performance metrics such as natural frequency, quality factor, mechanical and capacitive sensitivities, bandwidth, and pull-in voltage, revealing a significant influence on the device's performance. Moreover, through comparison with an accelerometer employing an equivalent uniform rectangular suspension spring, our findings underscore the superior flexibility of the hexagonal suspension springs for measuring low-g applications. These insights provide valuable guidance for the optimization of future MEMS accelerometer designs, catering to a wide range of application requirements.

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